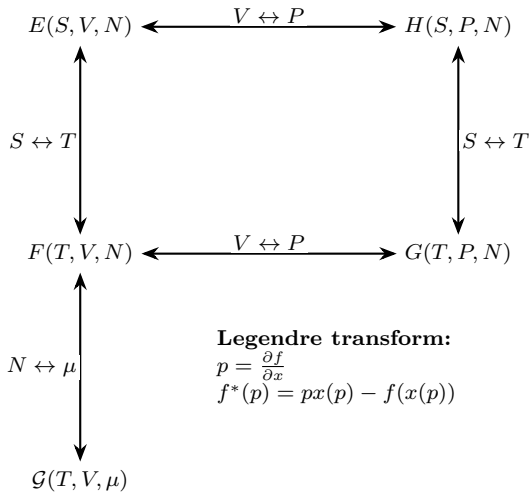


Statistical Mechanics — Formula Sheet

Exam 77802 A, 2026

Thermodynamics and potentials

Thermodynamic potentials



Legendre transform:

$$p = \frac{\partial f}{\partial x}$$

$$f^*(p) = px(p) - f(x(p))$$

$$\begin{aligned} E &= TS - PV + \mu N & dE &= TdS - PdV + \mu dN \\ H &= E + PV = TS + \mu N & dH &= TdS + VdP + \mu dN \\ F &= E - TS = -PV + \mu N & dF &= -SdT - PdV + \mu dN \\ G &= E - TS + PV = \mu N & dG &= -SdT + VdP + \mu dN \\ \mathcal{G} &= F - \mu N = -PV & d\mathcal{G} &= -SdT - PdV - Nd\mu \end{aligned}$$

Processes: isothermal $dT = 0$; isochoric $dV = 0$; isobaric $dP = 0$; adiabatic $\delta Q = 0$; isentropic $dS = 0$. For a reversible path $dS = \delta Q_{\text{rev}}/T$; generally $dS \geq \delta Q/T$. A reversible adiabatic process is isentropic.

Thermodynamic quantities

$$\beta = \frac{1}{k_B T}, \quad \frac{\partial \beta}{\partial T} = -\frac{1}{T^2 k_B} = -\beta^2 k_B, \quad C = \frac{\partial U}{\partial T}.$$

$$J_i \equiv \frac{\partial E(S, x_i)}{\partial x_i}, \quad dE = T dS + \sum_i J_i dx_i.$$

$$E(S, x_i) = T(S, x_i)S + \sum_i J_i x_i, \quad S dT + \sum_i x_i dJ_i = 0.$$

Entropy and maximum entropy

Boltzmann: for an isolated macrostate with multiplicity Ω ,

$$S_B = k_B \ln \Omega, \quad \frac{1}{T} = \left(\frac{\partial S}{\partial E} \right)_{V, N}.$$

Gibbs–Shannon:

$$S[p] = -k_B \sum_i p_i \ln p_i, \quad \sum_i p_i = 1.$$

Maximizing $S[p]$ subject to $\langle A_a \rangle = \sum_i p_i A_{a,i}$ gives

$$p_i = \frac{1}{Z} \exp \left(- \sum_a \lambda_a A_{a,i} \right), \quad Z = \sum_i \exp \left(- \sum_a \lambda_a A_{a,i} \right).$$

Canonical constraint $\langle E \rangle$: $\lambda_E = \beta = 1/(k_B T)$ and $p_i = Z^{-1} e^{-\beta E_i}$. Equal probabilities apply in the *isolated total system*; tracing out a heat bath produces the Boltzmann factor $\Omega_B(E_{\text{tot}} - E_i) \propto e^{-\beta E_i}$.

Transforms and ensemble choice

Legendre transform: if $p = f'(x)$, $f^*(p) = px(p) - f(x(p))$. It exchanges a controlled extensive variable with its intensive conjugate.
Laplace transform:

$$\mathcal{L}\{g\}(s) = \int_0^\infty e^{-sx} g(x) dx.$$

Canonical $Z(\beta)$ is the Laplace transform of the density of states: $Z(\beta) = \int \rho(E) e^{-\beta E} dE$.

z -transform / generating function:

$$\tilde{a}(z) = \sum_{N \geq 0} a_N z^N, \quad Z_G(z) = \sum_{N \geq 0} z^N Z_F, \quad z = e^{\beta \mu}.$$

Mostly used in the context of cluster expansion.

fixed variables	ensemble	potential
E, V, N	microcanonical	E or S
T, V, N	canonical	F
T, P, N	isothermal–isobaric	G
T, V, μ	grand canonical	Ω

Partition functions and derivatives

Canonical ensemble (T, V, N)

$$Z_F = \sum_i e^{-\beta E_i}, \quad F = -k_B T \ln Z_F, \quad U = -\partial_\beta \ln Z_F,$$

$$S = k_B (\ln Z_F + \beta U), \quad P = k_B T \partial_V \ln Z_F,$$

$$\text{Var}(E) = \partial_\beta^2 \ln Z_F = k_B T^2 C_V.$$

Classical phase space:

$$Z_F = \frac{1}{N! h^{3N}} \int \prod_{i=1}^N d^3 p_i d^3 q_i e^{-\beta H_N}.$$

Gibbs ensemble (T, P, N)

$$Z_G(\beta, P, N) = \int_0^\infty dV e^{-\beta PV} Z_F(\beta, V), \quad G = -k_B T \ln Z_G.$$

Here the effective energy is the enthalpy (Laplace transform of Z_F in V)

$$H = E + PV$$

$$\Rightarrow \langle H \rangle = -\partial_\beta \ln Z_G, \quad \langle V \rangle = -\frac{1}{\beta} \partial_P \ln Z_G,$$

$$S = k_B \ln Z_G + k_B \beta \langle H \rangle.$$

For spins in a magnetic field,

$$B \leftrightarrow -P, \quad M = \sum_i s_i \leftrightarrow V, \quad -BM \leftrightarrow PV.$$

Grand-canonical ensemble (T, V, μ)

$$Z_G(\beta, V, \mu) = \sum_{N=0}^\infty e^{\beta \mu N} Z_F(\beta, V, N) = \sum_N z^N Z_F(\beta, V, N)$$

The grand potential is

$$\mathcal{G} = -\frac{1}{\beta} \ln Z_G$$

The mean particle number and its fluctuations are

$$\langle N \rangle = \frac{1}{\beta} \partial_\mu \ln Z_G = z \partial_z \ln Z_G,$$

$$\text{Var}(N) = \frac{1}{\beta^2} \partial_\mu^2 \ln Z_G.$$

The entropy is

$$S = k_B \ln Z_G + k_B \beta (\langle E \rangle - \mu \langle N \rangle).$$

Ideal gas and equipartition

Thermal wavelength $\lambda_T = h/\sqrt{2\pi m k_B T}$:

$$Z_F = \frac{1}{N!} \left(\frac{V}{\lambda_T^3} \right)^N, \quad PV = N k_B T, \quad U = \frac{3}{2} N k_B T.$$

Using Stirling, $\ln N! \simeq N \ln N - N$:

$$F = -N k_B T \left[\ln \left(\frac{V}{N \lambda_T^3} \right) + 1 \right], \quad S = N k_B \left[\ln \left(\frac{V}{N \lambda_T^3} \right) + \frac{5}{2} \right].$$

Every independent quadratic term in H contributes $\frac{1}{2} k_B T$ to U and $\frac{1}{2} k_B$ to C_V .

Laplace / saddle-point method

For $N \gg 1$, $e^{-N f(x)}$ is dominated by the minimum x_* of f :

$$I(N) = \int g(x) e^{-N f(x)} dx \simeq g(x_*) e^{-N f(x_*)} \sqrt{\frac{2\pi}{N f''(x_*)}}.$$

In d dimensions, with Hessian $H_{ij} = \partial_i \partial_j f(x_*)$,

$$I(N) \simeq g(x_*) e^{-Nf(x_*)} \left(\frac{2\pi}{N} \right)^{d/2} (\det H)^{-1/2}.$$

Width $\Delta x \sim [Nf''(x_*)]^{-1/2}$; small parameter $1/N$ (or β at low T). Boundary / degenerate minima need a modified expansion.
Algorithm (low T , central potential). Want

$$I(\beta) = \int_{\mathbb{R}^3} e^{-\beta U(|\mathbf{q}|)} d^3 q, \quad \beta \gg 1,$$

with a minimum of U at q_* .

1. **Radial reduction:**

$$I(\beta) = 4\pi \int_0^\infty q^2 e^{-\beta U(q)} dq.$$

2. **Taylor of U about q_* :**

$$U(q) = U_0 + U_2(q - q_*)^2 + U_3(q - q_*)^3 + U_4(q - q_*)^4 + \dots,$$

with $U_0 = U(q_*)$, $U_2 = \frac{1}{2}U''(q_*)$, etc.

3. **Saddle:** set $x = q - q_*$; for $\beta \gg 1$ only $x \sim \beta^{-1/2}$ contributes,

$$I(\beta) \approx 4\pi e^{-\beta U_0} \int_{-\infty}^\infty (x + q_*)^2 \exp(-\beta U_2 x^2 - \beta U_3 x^3 - \dots) dx.$$

4. **Gaussian moments + corrections:**

$$\int_{-\infty}^\infty x^n e^{-ax^2} dx = \begin{cases} \frac{(n-1)!!}{(2a)^{n/2}} \sqrt{\frac{\pi}{a}} & n \text{ even,} \\ 0 & n \text{ odd.} \end{cases}$$

5. **Result:**

$$I(\beta) \approx 4\pi^{3/2} e^{-\beta U_0} q_*^2 \xi (1 + A_2 \xi^2 + A_4 \xi^4 + \dots),$$

with $\xi = 1/(\beta\sqrt{U_2})$ and $A_{2,4}$ fixed by higher derivatives of U at q_* .

Stirling: $N! = \int_0^\infty x^N e^{-x} dx$ gives

$$N! \simeq \sqrt{2\pi N} (N/e)^N \left(1 + \frac{1}{12N} + \dots \right).$$

Interactions, Mean Field and Phase Transitions

Transfer matrix: periodic 1D Ising model

$$H = -J \sum_{i=1}^N s_i s_{i+1} - h \sum_{i=1}^N s_i, \quad s_i = \pm 1, \quad s_{N+1} = s_1.$$

Split the field equally between adjacent bonds:

$$T_{ss'} = \exp \left[\beta J s s' + \frac{\beta h}{2} (s + s') \right], \quad T = \begin{pmatrix} e^{\beta J + \beta h} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J - \beta h} \end{pmatrix}.$$

$$Z_N = \text{Tr } T^N = \lambda_+^N + \lambda_-^N,$$

$$\lambda_{\pm} = e^{\beta J} \cosh(\beta h) \pm \sqrt{e^{2\beta J} \sinh^2(\beta h) + e^{-2\beta J}}.$$

For $N \rightarrow \infty$, $f = -k_B T \ln \lambda_+$ and $m = \partial_{\beta h} \ln \lambda_+$. Insert the diagonal spin operator $S = \text{diag}(1, -1)$:

$$\langle s_0 s_r \rangle = \frac{\text{Tr}(S T^r S T^{N-r})}{\text{Tr } T^N} = \left(\frac{\lambda_-}{\lambda_+} \right)^r + \left(\frac{\lambda_+}{\lambda_-} \right)^{N-r}.$$

At $h = 0$ and $N \rightarrow \infty$:

$$\langle s_0 s_r \rangle = (\tanh \beta J)^r = e^{-r/\xi}, \quad \xi^{-1} = -\ln(\tanh \beta J) = \ln \left| \frac{\lambda_0}{\lambda_1} \right|.$$

A phase transition would require correlations to persist over arbitrarily large distances, i.e. $\xi \rightarrow \infty$. In the 1D Ising model, $|\lambda_1| < \lambda_0$ for every $T > 0$, so ξ remains finite and there is no finite-temperature transition. Generally, the subleading eigenvalue sets the slowest-decaying correction and therefore controls long-range correlations.

Ising mean-field theory

Consider a system with nearest-neighbor interactions,

$$H_{\text{int}} = -J \sum_{\langle i,j \rangle} s_i s_j.$$

Define the fluctuation around the mean magnetization by

$$\delta s_i = s_i - m.$$

Let d be the spatial dimension and $z = 2d$ the coordination number, i.e. the number of nearest neighbors of each site.

Partition function:

$$Z_G = \text{tr} \exp \left\{ \beta \left(J \sum_{\langle i,j \rangle} s_i s_j + B \sum_i s_i \right) \right\}.$$

Using $s_i = m + \delta s_i$, using $[\dots] = \delta s_i \delta s_j + m(\delta s_i + \delta s_j) + m^2$

$$Z_G = \text{tr} \exp \left\{ \beta J \sum_{\langle i,j \rangle} [\dots] + \beta B \sum_i \delta s_i + \beta B N m \right\}.$$

Mean-field assumption:

$$\delta s_i \delta s_j \approx 0, \quad s_i s_j \approx m(s_i + s_j) - m^2.$$

Approximate partition function: Since

$$\sum_{\langle i,j \rangle} 1 = \frac{Nz}{2},$$

we obtain

$$Z_G^{\text{MF}} = \exp \left(-\frac{\beta J N z m^2}{2} \right) [2 \cosh(\beta B_{\text{MF}})]^N,$$

where

$$B_{\text{MF}} = B + J z m.$$

The self-consistency equation for the magnetization is

$$m = \frac{1}{\beta N} \frac{\partial \ln Z_G}{\partial B} = \tanh[\beta(B + J z m)].$$

Power-laws At $h = 0$, $T_c = zJ/k_B$. Near T_c :

$$m \sim (-t)^{1/2}, \quad \chi \sim |t|^{-1}, \quad m(T_c, h) \sim h^{1/3}, \quad t = (T - T_c)/T_c.$$

Hubbard–Stratonovich transformation

For $a > 0$,

$$e^{\frac{a}{2} x^2} = \frac{1}{\sqrt{2\pi a}} \int_{-\infty}^\infty d\phi \exp \left[-\frac{\phi^2}{2a} + \phi x \right].$$

For a positive matrix K ,

$$e^{\frac{1}{2} \mathbf{s}^T K \mathbf{s}} = \frac{1}{\sqrt{\det(2\pi K)}} \int d^n \phi e^{-\frac{1}{2} \phi^T K^{-1} \phi + \phi^T \mathbf{s}}.$$

The Hubbard–Stratonovich transformation trades a direct quadratic interaction between the microscopic variables for a linear interaction with a new auxiliary field. The new field ϕ is continuous and is integrated over all real values.

After the microscopic variables are summed or integrated out, ϕ acquires an effective action and describes fluctuations of the order parameter. Approximating the remaining integral by its dominant saddle point fixes ϕ to its most probable value and reproduces the mean-field approximation.

Landau and Ginzburg–Landau theory

Landau theory uses a coarse-grained order parameter φ (e.g. magnetization). (After HS, integrating out the microscopic degrees of freedom turns the auxiliary field ϕ into this Landau order parameter φ .) Above T_c , symmetry favors $\varphi = 0$; below T_c the system may choose $\varphi \neq 0$. Expand the free-energy density about $\varphi = 0$:

$$f(T, \varphi) = f_0(T) + \frac{1}{2}a(T)\varphi^2 + \frac{1}{4}u\varphi^4 - h\varphi + \dots,$$

with $u > 0$ and $a(T) \approx a_0(T - T_c) = a_0 T_c t$, $t = (T - T_c)/T_c$, $a_0 > 0$. Equivalently,

$$f(\varphi) = f_0 + \frac{a_0 t}{2}\varphi^2 + \frac{u}{4}\varphi^4 - h\varphi + \dots.$$

Minimize: at $h = 0$,

$$\frac{\partial f}{\partial \varphi} = 0 \implies a(T)\varphi + u\varphi^3 = 0.$$

Solutions: $\varphi = 0$ always, and $\varphi = \pm\sqrt{-a(T)/u}$ when $a(T) < 0$ (i.e.

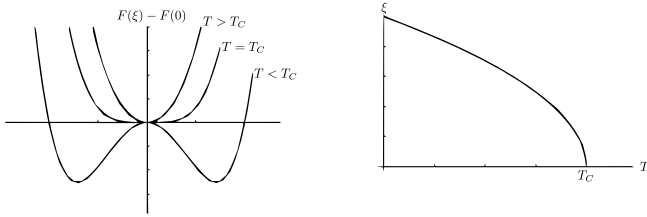
$T < T_c$). Near T_c ,

$$\varphi(T) = \pm\sqrt{\frac{a_0(T_c - T)}{u}} \sim (-t)^{1/2}, \quad \Rightarrow \quad \beta_{\text{MF}} = \frac{1}{2}.$$

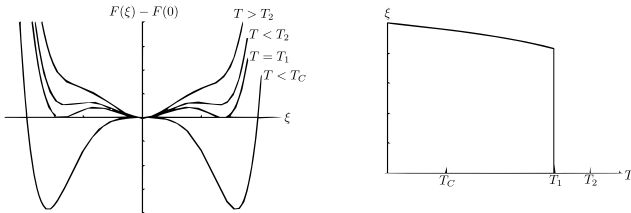
If $u > 0$ the order parameter grows continuously from zero (**second-order** / continuous transition). If the quartic coefficient is negative, a stabilizing $+v\varphi^6$ term is needed and the transition is first order; $u = 0$ is the tricritical point.

Phase classification (order parameter ξ):

Second Order Phase Transition



First Order Phase Transition



Ginzburg–Landau allows φ to vary in space:

$$\mathcal{F}[\varphi] = \int d^d x \left[\frac{c}{2}(\nabla\varphi)^2 + \frac{r}{2}\varphi^2 + \frac{u}{4}\varphi^4 - h\varphi \right], \quad Z = \int \mathcal{D}\varphi e^{-\beta\mathcal{F}[\varphi]}.$$

The gradient penalizes rapid spatial changes; the remaining terms set the locally preferred φ . Minimizing $\mathcal{F}$ gives

$$-c\nabla^2\varphi + r\varphi + u\varphi^3 = h.$$

Valid near a continuous transition for slow variations on scales $\gg$ lattice spacing (long-wavelength order-parameter fluctuations).

Critical exponents, scaling and universality

Near a continuous transition, physical quantities follow power laws:

$$C \sim |t|^{-\alpha}, \quad m \sim (-t)^\beta, \quad \chi \sim |t|^{-\gamma}, \quad m(t=0, h) \sim h^{1/\delta},$$

$$\xi \sim |t|^{-\nu}, \quad G(r, t=0) \sim r^{-(d-2+\eta)}.$$

These powers define the critical exponents. They appear because, as $t \rightarrow 0$, the correlation length diverges and the system becomes approximately scale invariant. Widom's scaling hypothesis assumes that the singular part of the free-energy density has the form

$$f_s(t, h) = |t|^{2-\alpha}\mathcal{F}_\pm(h/|t|^{\beta\delta}).$$

Thus t and h do not enter independently, but only through a scale-free combination. Taking derivatives of f_s , for example

$$m = -\partial_h f_s, \quad \chi = \partial_h m,$$

produces the power laws and relations between their exponents:

$$m = |t|^\beta \mathcal{M}_\pm(h/|t|^{\beta\delta}),$$

$$\alpha + 2\beta + \gamma = 2, \quad \gamma = \beta(\delta - 1), \quad \gamma = \nu(2 - \eta), \quad 2 - \alpha = d\nu.$$

Mean-field values for scalar ϕ^4 theory are (see 3 boxes above)

$$(\alpha, \beta, \gamma, \delta, \nu, \eta) = (0, \frac{1}{2}, 1, 3, \frac{1}{2}, 0).$$

Universality means that the exponents depend mainly on spatial dimension, symmetry, number of order-parameter components and interaction range, rather than on microscopic details.

Fourier-space intuition

For a lattice field s_i at positions $\mathbf{r}_i$,

$$s_i = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} s_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_i}, \quad s_{\mathbf{k}} = \frac{1}{\sqrt{N}} \sum_i s_i e^{-i\mathbf{k} \cdot \mathbf{r}_i}.$$

Translation-invariant quadratic forms diagonalize:

$$\frac{1}{2} \sum_{ij} s_i K_{i-j} s_j = \frac{1}{2} \sum_{\mathbf{k}} K(\mathbf{k}) |s_{\mathbf{k}}|^2.$$

Near a transition the soft modes have $ka \ll 1$, so $K(\mathbf{k}) \simeq r + ck^2 + \dots$; inverse Fourier transformation gives the Ginzburg–Landau gradient term.

Correlations, Goldstone Modes and Interacting Gases

Gaussian functional integrals and Green functions

For a positive operator K and dimensionless source J ,

$$Z[J] = \int \mathcal{D}\phi \exp \left[-\frac{1}{2} \phi K \phi + J\phi \right] = Z[0] \exp \left(\frac{1}{2} J K^{-1} J \right).$$

$$\langle \phi(x) \rangle_J = \frac{\delta \ln Z}{\delta J(x)},$$

$$G_c(x, y) = \langle \phi(x) \phi(y) \rangle_J = \frac{\delta^2 \ln Z}{\delta J(x) \delta J(y)} = K^{-1}(x, y).$$

For the Gaussian Ginzburg–Landau theory,

$$K(\mathbf{k}) = r + ck^2, \quad \tilde{G}(\mathbf{k}) = \frac{1}{r + ck^2}, \quad \xi = \sqrt{c/r}.$$

At large $R = |\mathbf{x}|$,

$$G(R) \sim \frac{e^{-R/\xi}}{R^{(d-1)/2}} \quad (R \gg \xi), \quad G(R) \sim R^{-(d-2)} \quad (r=0, d>2).$$

If the physical field couples as $-h \int \phi$, then

$$\chi = \frac{\partial \langle \phi \rangle}{\partial h} = \beta \int d^d x G_c(x, 0) = \beta \tilde{G}(0).$$

When $J = \beta h$ is used as the source, the same statement is $\partial \langle \phi \rangle / \partial J = \tilde{G}(0)$.

Goldstone modes

An $O(n)$ field is an n -component real field, $\vec{\phi}(\vec{x}) = (\phi_1(\vec{x}), \dots, \phi_n(\vec{x}))$, whose free energy is invariant under rotations $R \in O(n)$. For an $O(n)$ field,

$$\mathcal{F} = \int d^d x \left[\frac{c}{2} (\nabla \phi)^2 + \frac{r}{2} \phi^2 + \frac{u}{4} (\phi^2)^2 \right], \quad r < 0.$$

The minima form a continuous sphere of equivalent ordered states. Spontaneous symmetry breaking means that the system chooses one direction, for example

$$\langle \phi \rangle = v \hat{e}_1, \quad v^2 = -r/u,$$

and we decompose

$$\phi = (v + \sigma, \pi_1, \dots, \pi_{n-1}).$$

The longitudinal fluctuation σ changes the magnitude of the order parameter and moves away from the minimum, so it costs local energy. The transverse fluctuations π_a rotate the order parameter between equivalent minima, so a uniform rotation costs no energy:

$$m_L^2 = 2|r| \quad (\sigma, \text{longitudinal}), \quad m_T^2 = 0 \quad (\pi_a, \text{transverse}).$$

Thus spontaneous breaking $G \rightarrow H$ creates $\dim G - \dim H$ massless Goldstone modes: one for each independent direction along the

degenerate vacuum manifold. For $O(n) \rightarrow O(n-1)$, there are $n-1$ such modes, with

$$G_T(k) \sim \frac{1}{ck^2}.$$

Only spatial variations of these rotations cost gradient energy, which vanishes for sufficiently long wavelengths.

Upper/lower critical dimensions and Ginzburg criterion

The lower critical dimension asks: in how few dimensions can an ordered phase survive fluctuations? For a system with a broken continuous symmetry, the massless Goldstone modes have

$$G_T(k) \sim \frac{1}{k^2}, \quad \langle \pi^2 \rangle \sim \int_{k_{\min}} \frac{d^d k}{k^2}.$$

For $d \leq 2$, the long-wavelength contribution $k \rightarrow 0$ diverges. Large-scale rotations then destroy the chosen direction of the order parameter, so there is no finite- T continuous symmetry breaking in short-range systems (Mermin–Wagner). Hence $d_\ell = 2$ for $O(n \geq 2)$. For the discrete Ising symmetry there are no Goldstone modes, and $d_\ell = 1$.

The upper critical dimension asks: above which dimension are fluctuations weak enough that mean-field theory becomes correct? For ϕ^4 ,

$$g(\xi) \sim \frac{uk_B T}{c^2} \xi^{4-d}$$

measures the strength of fluctuations on the correlation-length scale. Mean field is self-consistent when $g(\xi) \ll 1$ (Ginzburg criterion). Since $\xi \rightarrow \infty$ near the transition, fluctuations grow for $d < 4$, are marginal at $d = 4$, and become irrelevant for $d > 4$. Therefore $d_c = 4$: below it the exponents are non-mean-field, at it logarithmic corrections appear, and above it mean-field theory is asymptotically correct. Hyperscaling generally fails above d_c .

Mayer expansion and linked clusters

For pair interactions $U = \sum_{i < j} u(r_{ij})$ define

$$f_{ij} = e^{-\beta u(r_{ij})} - 1, \quad e^{-\beta U} = \prod_{i < j} (1 + f_{ij}).$$

Each term corresponds to a graph: particles are vertices and f_{ij} are edges. The configurational integral is a sum over all graphs. Every graph factorizes into connected components; exponentiating the connected-cluster sum gives

$$\ln Z_G = V \sum_{\ell \geq 1} b_\ell z^\ell, \quad \beta P = \sum_{\ell \geq 1} b_\ell z^\ell, \quad n = \sum_{\ell \geq 1} \ell b_\ell z^\ell,$$

where this convention absorbs the powers of λT into z, b_ℓ . The cluster coefficients keep only *connected* graphs $\mathcal{C}_\ell$:

$$b_\ell = \frac{1}{\ell! V} \int_{V^\ell} \left(\sum_{G \in \mathcal{C}_\ell} \prod_{(i,j) \in G} f_{ij} \right) \prod_{i=1}^{\ell} d^d r_i.$$

The logarithm drops disconnected graphs because they factorize into products already generated by the exponential.

Example $\ell = 2$: only one connected graph, a single bond f_{12} :

$$b_2 = \frac{1}{2!V} \int f_{12} d^d r_1 d^d r_2 = \frac{1}{2} \int d^d r f(r).$$

$\ell = 3$: The connected graphs are the three open chains and the triangle:

$$\sum_{G \in \mathcal{C}_3} \prod f_{ij} = f_{12}f_{13} + f_{12}f_{23} + f_{13}f_{23} + f_{12}f_{13}f_{23}.$$

Hence

$$b_3 = \frac{1}{3!V} \int_{V^3} [f_{12}f_{13} + f_{12}f_{23} + f_{13}f_{23} + f_{12}f_{13}f_{23}] \prod_{i=1}^3 d^d r_i.$$

The open chains are reducible: removing their central particle disconnects the graph. The triangle remains connected after removing any one particle and is therefore irreducible. Connected graphs determine $\ln Z_G$, while the virial coefficients contain only irreducible graphs; consequently, only the triangle contributes directly to B_3 . For a partition with n_k clusters of size k ,

$$\sum_k k n_k = N, \quad \Omega_N(\{n_k\}) = \frac{N!}{\prod_k (k!)^{n_k} n_k!}.$$

Small parameter: high- T interactions ($|f| \ll 1$).

Virial expansion and basic coefficients

The grand-canonical calculation naturally gives pressure and density as functions of the fugacity z . Eliminating z between $P(z)$ and $n(z)$ gives the virial expansion,

$$\beta P = n + B_2 n^2 + B_3 n^3 + \dots$$

The first term is the ideal-gas law; the higher powers describe corrections due to interactions between pairs, triples, and larger groups of particles. If

$$\beta P = z + b_2 z^2 + b_3 z^3 + \dots,$$

then

$$B_2 = -b_2, \quad B_3 = 4b_2^2 - 2b_3.$$

For a translation-invariant pair potential,

$$B_2 = -\frac{1}{2} \int d^d r f(r),$$

$$B_3 = -\frac{1}{3} \int d^d r_{12} d^d r_{13} f_{12} f_{13} f_{23}.$$

Thus B_2 measures the effect of one interacting pair, while B_3 contains the genuinely three-particle contribution in which all three particles are linked. The virial expansion is useful experimentally because density is measurable, whereas fugacity is not.

For hard spheres of diameter σ in three dimensions,

$$B_2 = \frac{2\pi}{3} \sigma^3, \quad B_3 = \frac{5\pi^2}{18} \sigma^6 = \frac{5}{8} B_2^2.$$

Here B_2 is half the volume excluded to the center of a second sphere. More generally, for a d -dimensional hard core,

$$B_2 = \frac{1}{2} V_d(\sigma), \quad V_d(R) = \frac{\pi^{d/2} R^d}{\Gamma(d/2 + 1)}.$$

Perturbative-expansion checklist

- **Laplace/low- T :** expand about the dominant minimum; small width $\Delta x \sim N^{-1/2}$ or $\beta^{-1/2}$.
- **Mayer/cluster:** expand $\prod (1 + f_{ij})$; controlled by weak bonds/high T and low density.
- **Virial:** expand in nR^d ; invert the fugacity series carefully.
- **Landau:** expand in a small order parameter and long gradients near a continuous transition; retain all symmetry-allowed terms needed for stability.

Stochastic Processes, Noise and Path Integrals

Markov processes and Chapman–Kolmogorov

A process is Markovian when the future conditioned on the present is independent of the past:

$$P(x_n, t_n | x_{n-1}, t_{n-1}; \dots; x_0, t_0) = P(x_n, t_n | x_{n-1}, t_{n-1}).$$

Transition kernels obey Chapman–Kolmogorov:

$$P(x_2, t_2 | x_0, t_0) = \int dx_1 P(x_2, t_2 | x_1, t_1) P(x_1, t_1 | x_0, t_0).$$

For a homogeneous discrete chain, $\mathbf{P}_{n+1} = T\mathbf{P}_n$, $\sum_i T_{ij} = 1$, where T_{ij} is the transition probability from state i to state j . A stationary distribution is an eigenvector with eigenvalue 1; modes with $|\lambda| < 1$ decay. Brownian motion is approximately Markovian when molecular collision times are much shorter than the observed time scale.

Wiener process and Gaussian white noise

A Wiener process $W(t)$ has $W(0) = 0$, continuous paths, independent stationary increments, and

$$W(t + \Delta t) - W(t) \sim \mathcal{N}(0, \Delta t), \quad \langle W(t) \rangle = 0, \\ \langle W(t)W(s) \rangle = \min(t, s).$$

Gaussian white noise is the distributional derivative $\eta(t) = \dot{W}(t)$:

$$\langle \eta(t) \rangle = 0, \quad \langle \eta(t)\eta(t') \rangle = \delta(t - t').$$

For strength Λ , write $\xi(t) = \sqrt{\Lambda} \eta(t)$, so $\langle \xi(t)\xi(t') \rangle = \Lambda \delta(t - t')$. On a grid,

$$\langle \xi_i \xi_j \rangle = \frac{\Lambda}{\Delta t} \delta_{ij}, \quad \mathcal{P}[\xi] \propto \exp \left[-\frac{1}{2\Lambda} \int \xi(t)^2 dt \right].$$

Multivariate Gaussian identities

For symmetric positive A and mean zero,

$$p(\mathbf{x}) = \frac{\sqrt{\det A}}{(2\pi)^{n/2}} \exp \left(-\frac{1}{2} \mathbf{x}^T A \mathbf{x} \right), \quad C = A^{-1}.$$

$$I(\mathbf{b}) = \langle e^{\mathbf{b}^T \mathbf{X}} \rangle = \exp \left(\frac{1}{2} \mathbf{b}^T C \mathbf{b} \right), \quad \langle X_i X_j \rangle = C_{ij}.$$

All odd centered moments vanish. Wick/Isserlis theorem:

$$\langle X_1 X_2 X_3 X_4 \rangle = C_{12} C_{34} + C_{13} C_{24} + C_{14} C_{23}.$$

For Gaussian variables, zero covariance implies independence.

Langevin equations and moments

Underdamped Brownian particle:

$$m\dot{v} = -\gamma v + F(x) + \xi(t), \quad \langle \xi(t)\xi(t') \rangle = 2\gamma k_B T \delta(t - t').$$

For $F = 0$, $\tau_B = m/\gamma$ and

$$v(t) = v_0 e^{-t/\tau_B} + \frac{1}{m} \int_0^t e^{-(t-s)/\tau_B} \xi(s) ds,$$

$$\langle v \rangle = v_0 e^{-t/\tau_B}, \quad \text{Var}(v) = \frac{k_B T}{m} (1 - e^{-2t/\tau_B}).$$

Long times: $\langle [x(t) - x(0)]^2 \rangle \simeq 2Dt$ in one dimension, $D = k_B T/\gamma$. Overdamped form:

$$dx = a(x, t) dt + b(x, t) dW_t.$$

For Itô calculus,

$$\frac{d}{dt} \langle A(x) \rangle = \left\langle aA' + \frac{1}{2} b^2 A'' \right\rangle.$$

Taking $A = x, x^2, \dots$ gives moment equations without solving the full PDF.

Gaussian path integrals

A path integral is an ordinary multidimensional integral with one variable for each time or spatial point:

$$\int \mathcal{D}\phi \sim \lim_{N \rightarrow \infty} \int \prod_{i=1}^N d\phi_i.$$

For an ordinary Gaussian integral,

$$\int d^n x e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x} + \mathbf{J}^T \mathbf{x}} = \sqrt{\frac{(2\pi)^n}{\det A}} e^{\frac{1}{2} \mathbf{J}^T A^{-1} \mathbf{J}}.$$

The functional analogue is

$$\int \mathcal{D}\phi e^{-\frac{1}{2} \int \phi K \phi + \int J \phi} \propto (\det K)^{-1/2} e^{\frac{1}{2} \int J K^{-1} J}.$$

Thus K^{-1} is the covariance, or Green's function, of the Gaussian field.

For the Langevin equation

$$\dot{x}(t) = a(x, t) + \xi(t), \quad \langle \xi(t)\xi(t') \rangle = \Lambda \delta(t - t'),$$

a trajectory $x(t)$ requires the noise realization

$$\xi(t) = \dot{x}(t) - a(x, t).$$

Since Gaussian white noise has probability

$$\mathcal{P}[\xi] \propto \exp \left[-\frac{1}{2\Lambda} \int \xi(t)^2 dt \right],$$

the corresponding trajectory probability is

$$\mathcal{P}[x] \propto \mathcal{J}[x] \exp \left[-\frac{1}{2\Lambda} \int (\dot{x}(t) - a(x, t))^2 dt \right].$$

Trajectories close to the deterministic equation

$$\dot{x}(t) = a(x, t)$$

are therefore the most probable.

A functional delta enforces an equation at every time:

$$\delta[F] \propto \int \mathcal{D}\hat{x} \exp \left[i \int \hat{x}(t) F(t) dt \right].$$

For example, choosing

$$F(t) = \dot{x}(t) - a(x, t) - \xi(t)$$

imposes the Langevin equation inside the path integral.

The Jacobian $\mathcal{J}[x]$ depends on the discretization convention and is often constant for additive or linear noise.

Kramers–Moyal expansion

Suppose that during a short time interval τ , a stochastic variable changes by

$$\Delta x = x(t + \tau) - x(t),$$

given that it started at $x(t) = x$.

The short-time motion can be characterized by its moments:

$$\langle \Delta x | x(t) = x \rangle, \quad \langle (\Delta x)^2 | x(t) = x \rangle, \quad \langle (\Delta x)^3 | x(t) = x \rangle, \dots$$

The Kramers–Moyal coefficients measure how quickly these moments grow:

$$D^{(n)}(x, t) = \lim_{\tau \rightarrow 0} \frac{1}{n! \tau} \langle (\Delta x)^n | x(t) = x \rangle.$$

In particular,

$$D^{(1)} = \lim_{\tau \rightarrow 0} \frac{\langle \Delta x \rangle}{\tau}$$

is the local average velocity, or drift, while

$$D^{(2)} = \lim_{\tau \rightarrow 0} \frac{\langle (\Delta x)^2 \rangle}{2\tau}$$

measures the local spreading due to randomness.

These short-time moments determine the evolution of the probability density:

$$\partial_t P(x, t) = \sum_{n=1}^{\infty} (-\partial_x)^n \left[D^{(n)}(x, t) P(x, t) \right].$$

If $D^{(n)} = 0$ for $n \geq 3$, this truncates to the **Fokker–Planck equation**

$$\partial_t P = -\partial_x (D^{(1)} P) + \partial_x^2 (D^{(2)} P).$$

For the Itô stochastic equation

$$dx = a(x, t) dt + b(x, t) dW,$$

the short-time change is

$$\Delta x \simeq a\tau + b\Delta W,$$

with

$$\langle \Delta W \rangle = 0, \quad \langle (\Delta W)^2 \rangle = \tau.$$

Therefore,

$$\langle \Delta x \rangle \simeq a\tau, \quad \langle (\Delta x)^2 \rangle \simeq b^2 \tau,$$

and all higher moments vanish after division by τ in the limit $\tau \rightarrow 0$. Hence,

$$D^{(1)} = a, \quad D^{(2)} = \frac{1}{2} b^2, \quad D^{(n)} = 0 \quad \text{for } n \geq 3.$$

Substituting these coefficients recovers

$$\partial_t P = -\partial_x(aP) + \frac{1}{2}\partial_x^2(b^2 P).$$

Thus, a transports probability, while b causes probability to spread.
Pawula theorem: A consistent finite truncation can contain either only the drift term, or the drift and diffusion terms. One cannot consistently stop at third or fourth order while preserving $P(x, t) \geq 0$.

Fokker–Planck, Master Equations and Monte Carlo

Fokker–Planck equation and probability current

For Itô $dx = a(x, t) dt + b(x, t) dW_t$,

$$\partial_t P = -\partial_x[aP] + \frac{1}{2}\partial_x^2[b^2 P] = -\partial_x J,$$

$$J = aP - \frac{1}{2}\partial_x(b^2 P).$$

In d dimensions, with $dx_i = a_i dt + B_{i\alpha} dW_\alpha$ and $D_{ij} = (BB^T)_{ij}$,

$$\partial_t P = -\partial_i(a_i P) + \frac{1}{2}\partial_i\partial_j(D_{ij} P).$$

Boundary conditions: reflecting $J \cdot \hat{n} = 0$; absorbing $P = 0$.
 Stationarity requires $\partial_x J = 0$; equilibrium with no net current has $J = 0$. For additive overdamped motion

$$dx = \frac{F(x)}{\gamma} dt + \sqrt{2D} dW, \quad D = \frac{k_B T}{\gamma},$$

$F = -U'$ gives

$$P_{\text{eq}}(x) = Z^{-1} e^{-\beta U(x)}.$$

Klein–Kramers equation

For

$$\dot{x} = v, \quad m\dot{v} = -\gamma v - U'(x) + \xi(t), \quad \langle \xi(t)\xi(t') \rangle = 2\gamma k_B T \delta(t - t'),$$

$$\partial_t P = -v\partial_x P + \frac{1}{m}U'(x)\partial_v P + \frac{\gamma}{m}\partial_v(vP) + \frac{\gamma k_B T}{m^2}\partial_v^2 P.$$

The stationary solution is Maxwell–Boltzmann:

$$P_{\text{eq}}(x, v) = Z^{-1} \exp\left[-\beta\left(\frac{1}{2}mv^2 + U(x)\right)\right].$$

In the strongly overdamped limit, eliminate v to obtain the Smoluchowski Fokker–Planck equation for x .

Continuous-time master equation

Let W_{ij} be the transition rate $j \rightarrow i$. Then

$$\dot{P}_i(t) = \sum_{j \neq i} [W_{ij}P_j - W_{ji}P_i] = \sum_j L_{ij}P_j,$$

where $L_{ij} = W_{ij}$ for $i \neq j$ and $L_{ii} = -\sum_{k \neq i} W_{ki}$, so $\sum_i L_{ij} = 0$. A stationary state satisfies $L\mathbf{P}^{\text{st}} = 0$. The zero eigenvalue is the stationary mode; the nonzero eigenvalues determine relaxation times. Multiple closed communicating classes may give multiple stationary states. Probability conservation follows by summing the master equation over i .

Detailed balance

Detailed balance with respect to P^{eq} means pairwise cancellation:

$$W_{ij}P_j^{\text{eq}} = W_{ji}P_i^{\text{eq}} \quad \forall i, j.$$

It is sufficient for stationarity but stronger than stationarity: a stationary nonequilibrium state may have circulating probability currents. For the canonical distribution,

$$\frac{W_{ij}}{W_{ji}} = \frac{P_i^{\text{eq}}}{P_j^{\text{eq}}} = e^{-\beta(E_i - E_j)}.$$

Kolmogorov cycle criterion (for positive rates): detailed balance holds iff the product of rates around every closed cycle equals the product around the reversed cycle.

Trigonometry

Angle addition formulas:

$$\sin(a \pm b) = \sin(a) \cos(b) \pm \cos(a) \sin(b)$$

$$\cos(a \pm b) = \cos(a) \cos(b) \mp \sin(a) \sin(b)$$

Product-to-sum formulas:

$$\cos(a) \cos(b) = \frac{1}{2} [\cos(a+b) + \cos(a-b)]$$

$$\sin(a) \sin(b) = \frac{1}{2} [\cos(a-b) - \cos(a+b)]$$

$$\cos(a) \sin(b) = \frac{1}{2} [\sin(a+b) + \sin(a-b)]$$

Using complex exponentials:

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i} \quad \cos(z) = \frac{e^{iz} + e^{-iz}}{2}$$

$$\cosh(x) = \cos(ix) \quad \sinh(x) = -i \sin(ix)$$

Double-angle identities:

$$\cos(2x) = \cos^2(x) - \sin^2(x) \quad \sin(2x) = 2 \sin(x) \cos(x)$$

Half-angle identities:

$$\sin\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos \theta}{2}} \quad \cos\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$\tan\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \frac{\sin \theta}{1 + \cos \theta} = \frac{1 - \cos \theta}{\sin \theta}$$

Powers of Trigonometric Functions

$$\cos^2(x) = \frac{1}{2} + \frac{1}{2} \cos(2x) \quad \sin^2(x) = \frac{1}{2} - \frac{1}{2} \cos(2x)$$

$$\cos^3(x) = \frac{3}{4} \cos(x) + \frac{1}{4} \cos(3x) \quad \sin^3(x) = \frac{3}{4} \sin(x) - \frac{1}{4} \sin(3x)$$

Functional Derivative

For $J[f] = \int_a^b \mathcal{L}(x, f(x), f'(x)) \, dx$, we obtain:

$$\frac{\delta J}{\delta f(x)} = \frac{\partial \mathcal{L}}{\partial f}(x) - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial f'}(x)$$

For $J[f] = \int \mathcal{L}(x, f(x)) \, dx$, we obtain:

$$\frac{\delta J}{\delta f(y)} = \frac{\partial \mathcal{L}}{\partial f}(x = y, f(y))$$

Spherical Coordinates

Spherical-coordinate parametrization:

$$x = r \sin \theta \cos \phi \quad y = r \sin \theta \sin \phi \quad z = r \cos \theta$$

Coordinate ranges:

$$r \geq 0, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \phi < 2\pi$$

Volume element:

$$d^3\mathbf{r} = r^2 \sin \theta \, dr \, d\theta \, d\phi$$

Integrals

$$\int x e^{ax} \, dx = \frac{e^{ax}}{a} \left(x - \frac{1}{a} \right) \quad \int x^2 e^{ax} \, dx = \frac{e^{ax}}{a} \left(x - \frac{1}{a} \right)$$

$$\int x \cos(ax) \, dx = \frac{\cos(ax)}{a^2} + \frac{x \sin(ax)}{a}$$

$$\int x \sin(ax) \, dx = \frac{\sin(ax)}{a^2} - \frac{x \cos(ax)}{a}$$

$$\int x^2 \cos(ax) \, dx = \frac{2x}{a^2} \cos(ax) + \left(\frac{x^2}{a} - \frac{2}{a^3} \right) \sin(ax)$$

$$\int x^2 \sin(ax) \, dx = \frac{2x}{a^2} \sin(ax) + \left(\frac{2}{a^3} - \frac{x^2}{a} \right) \cos(ax)$$

$$\int \sin(ax) \cos(bx) \, dx = -\frac{\cos[(a-b)x]}{2(a-b)} - \frac{\cos[(a+b)x]}{2(a+b)}$$

$$\int \cos(ax) \cos(bx) \, dx = \frac{\sin[(a-b)x]}{2(a-b)} + \frac{\sin[(a+b)x]}{2(a+b)}$$

$$\int \sin(ax) \sin(bx) \, dx = \frac{\sin[(a-b)x]}{2(a-b)} - \frac{\sin[(a+b)x]}{2(a+b)}$$

$$\int \sin^n(ax) \cos(ax) \, dx = \frac{\sin^{n+1}(ax)}{(n+1)a}$$

$$\int \cos^n(ax) \cos(ax) \, dx = -\frac{\cos^{n+1}(ax)}{(n+1)a}$$

$$\int e^{ax} \sin(bx) \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin(bx) - b \cos(bx))$$

$$\int e^{ax} \cos(bx) \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos(bx) + b \sin(bx))$$

Definite Integrals

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} \, dx = \sqrt{\frac{\pi}{\alpha}} \quad \int_{-\infty}^{\infty} e^{-\alpha x^2 + i\beta x} \, dx = \sqrt{\frac{\pi}{\alpha}} e^{-\frac{\beta^2}{4\alpha}}$$

$$\int_0^{\infty} e^{-ax} \cos(bx) \, dx = \frac{a}{a^2 + b^2} \quad \int_0^{\infty} e^{-ax} \sin(bx) \, dx = \frac{b}{a^2 + b^2}$$

$$\int_0^{\pi/2} \sin^2(x) \, dx = \int_0^{\pi/2} \cos^2(x) \, dx = \frac{\pi}{4} \quad \int_{-\pi/2}^{\pi/2} \cos(x) \, dx = 2$$

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(x) \, dx = f(b(x)) b'(x) - f(a(x)) a'(x)$$

Basic Differential Equations

$$\ddot{Y} + \lambda^2 Y = 0 \Rightarrow Y = A \sin(\lambda x) + B \cos(\lambda x)$$

$$\ddot{Y} - \lambda^2 Y = 0 \Rightarrow Y = A \sinh(\lambda x) + B \cosh(\lambda x) = C e^{\lambda x} + D e^{-\lambda x}$$

$$\dot{Y} + \lambda Y = 0 \Rightarrow Y = A e^{-\lambda x}$$

$$\ddot{Y} + \lambda = 0 \Rightarrow Y = -\frac{\lambda}{2} x^2 + Ax + B$$

$$r^2 R'' + r R' - \lambda^2 = 0 \Rightarrow R = A r^{\lambda} + B r^{-\lambda}$$

$$y' + P(x)y = Q(x) \Rightarrow y = e^{-\int P(x) \, dx} \left(\int e^{\int P(x) \, dx} Q(x) \, dx + C \right)$$